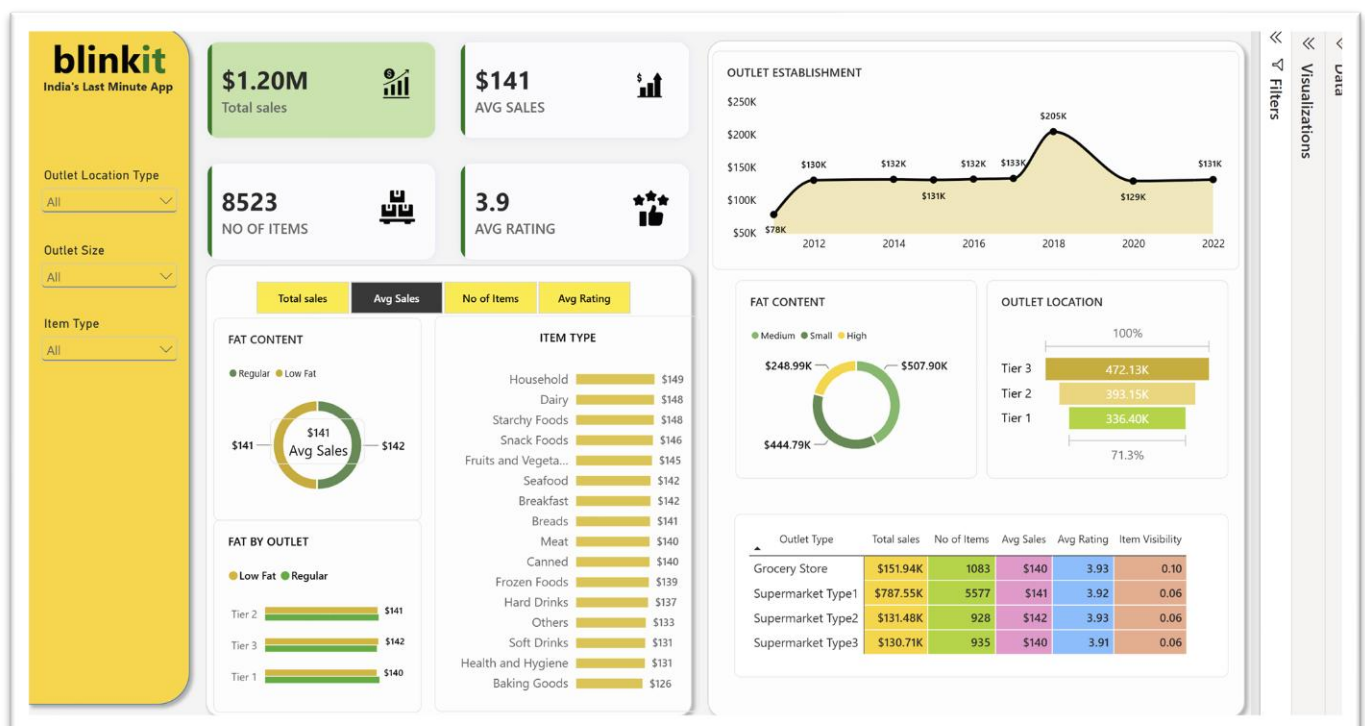


PROJECT REPORT

Blinkit Sales Analysis Dashboard

An Interactive Business Intelligence Project using Power BI

Project Domain	Sales & Business Intelligence
Primary Tool	Microsoft Power BI
Techniques	Power Query, DAX, Data Visualization
Dashboard Focus	Grocery Sales & Outlet Performance
Prepared By	Mayank Gupta



1. Executive Summary

This project presents an interactive Blinkit Sales Analysis Dashboard developed in Microsoft Power BI. The objective is to transform grocery sales data into a clear business intelligence solution that helps users understand sales performance, product behavior, outlet performance, outlet characteristics, and customer ratings. The dashboard combines key performance indicators, comparative charts, trend analysis, and interactive slicers to support fast exploration of the underlying business data.

2. Project Background

Blinkit is an online grocery delivery business where sales performance can be evaluated across products, fat-content categories, outlet types, outlet sizes, location tiers, and outlet establishment years. A dashboard provides a convenient way to consolidate these dimensions into a single analytical view. This project focuses on building a decision-support dashboard rather than presenting raw transactional data.

3. Problem Statement

Raw sales data can contain many product and outlet attributes, making it difficult to identify important patterns using spreadsheets or unstructured reports. The project addresses this problem by creating a Power BI dashboard that organizes the information into KPIs and visual analyses. The solution is intended to help users answer questions such as which outlet types perform better, how sales vary by product type, how outlet location affects performance, and how sales have changed over outlet establishment years.

4. Project Objectives

- Build an interactive sales analysis dashboard in Power BI.
- Track overall sales and other important business KPIs.
- Analyze sales by product type and item fat content.
- Compare outlet performance by location, size, and outlet type.
- Study sales trends based on outlet establishment year.
- Provide interactive filtering through slicers.
- Present business information in a professional and decision-friendly format.

5. Tools and Technologies

Tool / Technology	Purpose
Microsoft Power BI	Dashboard development, visualization and interactive reporting
Power Query	Data preparation, cleaning and transformation
DAX	Measures and analytical calculations
Excel / Tabular Data	Source data format and initial data inspection

6. Data Overview

The dashboard uses a grocery sales dataset containing product-level and outlet-level attributes. The model includes the table named "BlinkIT Grocery Data". Key analytical fields include item type, item fat content, item visibility, outlet establishment year, outlet size, outlet location type, outlet type, item sales and item ratings. These fields allow both product-level and outlet-level analysis.

7. Data Preparation and Transformation

- Imported the grocery sales data into Power BI.
- Reviewed field names, data types and categorical attributes.
- Prepared the dataset using Power Query where required.
- Ensured numerical fields were suitable for aggregation.
- Prepared dimensions such as item type, outlet size, location type and outlet type for analysis.
- Created analytical measures using DAX for dashboard KPIs.

8. Key Performance Indicators

KPI	Business Meaning
Total Sales	Overall sales generated across the available records
Average Sales	Average sales value used for comparative analysis
Number of Items	Count of items represented in the analysis
Average Rating	Average customer rating associated with the items

9. Dashboard Design and Visualizations

The dashboard is designed as a single-page analytical report. It combines KPI cards, categorical comparisons, trend analysis, outlet analysis and slicers. The layout is intended to let a user move from high-level business performance to detailed comparisons without leaving the dashboard.

Visual	Analysis
KPI Cards	Total Sales, Average Sales, Number of Items and Average Rating
FAT CONTENT Donut	Average sales comparison by item fat content
Location & Fat Content Chart	Average sales comparison across outlet location type and item fat content
Item Type Bar Chart	Average sales by product/item type
Establishment Year Line Chart	Total sales trend by outlet establishment year
Outlet Size Donut	Total sales distribution by outlet size
OUTLET LOCATION Funnel	Sales comparison by outlet location type
Outlet Type Matrix	Sales, items, average sales, average rating and average item visibility by outlet type
Slicers	Outlet location type, outlet size, item type and selectable metrics

10. DAX and Measure Strategy

DAX measures are used to calculate business metrics dynamically so that the dashboard responds to user filters. Typical measures in this project include total sales, average sales, item count and average rating. The dashboard also uses a metrics-selection concept to allow a user to change the analytical metric displayed in selected visuals. The exact measure definitions should be maintained in the PBIX model and can be documented separately for version control.

11. Interactivity and User Experience

- Slicers allow users to filter the dashboard by outlet location type, outlet size and item type.
- A metric selector provides an interactive way to change the selected analytical measure.
- Cross-filtering enables users to explore relationships between product and outlet dimensions.
- KPI cards provide immediate visibility into the overall performance of the filtered data.
- Multiple chart types are used to make comparisons, distributions and trends easier to interpret.

12. Analytical Questions Addressed

- What is the overall sales performance?
- How does average sales vary by item type?
- How do different fat-content categories contribute to average sales?
- How does outlet location type relate to sales performance?
- How are sales distributed across outlet sizes?
- How has total sales changed according to outlet establishment year?
- Which outlet types show stronger sales and item performance?
- How do outlet types differ in average rating and item visibility?

13. Business Insights Framework

The dashboard is structured to help identify high-performing and low-performing segments rather than relying on a single overall sales number. Users can compare product categories, outlet characteristics and location types to discover areas that may require attention. Because the dashboard is interactive, the same questions can be evaluated under different filters, supporting more focused business analysis.

14. Recommendations

- Monitor outlet types and location tiers that consistently underperform.
- Evaluate high-performing product categories for inventory and promotional opportunities.
- Use outlet-size comparisons to understand where sales concentration is strongest.
- Investigate changes in performance across establishment years to identify operational patterns.
- Use customer rating alongside sales rather than treating sales as the only performance indicator.
- Use interactive filtering during business reviews to investigate specific product and outlet segments.

15. Project Workflow

Step	Activity
1	Collect and inspect source grocery sales data
2	Clean and transform data using Power Query
3	Define fields and analytical dimensions
4	Create DAX measures for KPIs
5	Build charts, KPI cards and matrix
6	Add slicers and interactive metric selection
7	Format dashboard for professional presentation
8	Validate visuals and filter interactions
9	Prepare dashboard for portfolio/GitHub presentation

16. Project Deliverables

- Power BI dashboard file (.pbix).
- Interactive sales analysis report.
- Project documentation/report.
- Dashboard screenshot for portfolio or GitHub README.
- Optional DAX measure documentation.

17. Limitations

- The dashboard is dependent on the quality and completeness of the source dataset.
- The report provides descriptive and diagnostic analysis; it is not a forecasting model.
- Business conclusions should be validated against current operational data before making major decisions.
- The report does not establish causal relationships between outlet characteristics and sales.

18. Conclusion

The Blinkit Sales Analysis Dashboard demonstrates how Power BI can transform grocery sales data into an interactive business intelligence solution. By combining KPI cards, categorical comparisons, trend analysis, outlet-level analysis, DAX measures and interactive filters, the project provides a structured way to explore sales performance. The project also demonstrates practical skills in data preparation, analytical modeling, visualization and dashboard design, making it suitable as a portfolio project for a Data Analyst role.

19. Skills Demonstrated

- Data Cleaning and Transformation
- Power Query
- DAX Measures
- Data Visualization
- Business Intelligence
- KPI Development
- Interactive Dashboard Design
- Sales Performance Analysis
- Analytical Storytelling
- Portfolio Project Presentation

20. Portfolio / Repository Structure

```
Blinkit-Sales-Analysis-/  
■■■ README.md  
■■■ Dashboard/  
■ ■■■ BLINKIT_SALES_ANALYSIS.pbix  
■■■ Screenshots/  
■■■ blinkit_dashboard.png
```

The source dataset can be included only when its redistribution is permitted. The PBIX file is best kept inside the Dashboard folder, while a dashboard screenshot is useful for previewing the project on GitHub.

21. Author

Mayank Gupta

Data Analyst / Business Intelligence Portfolio Project